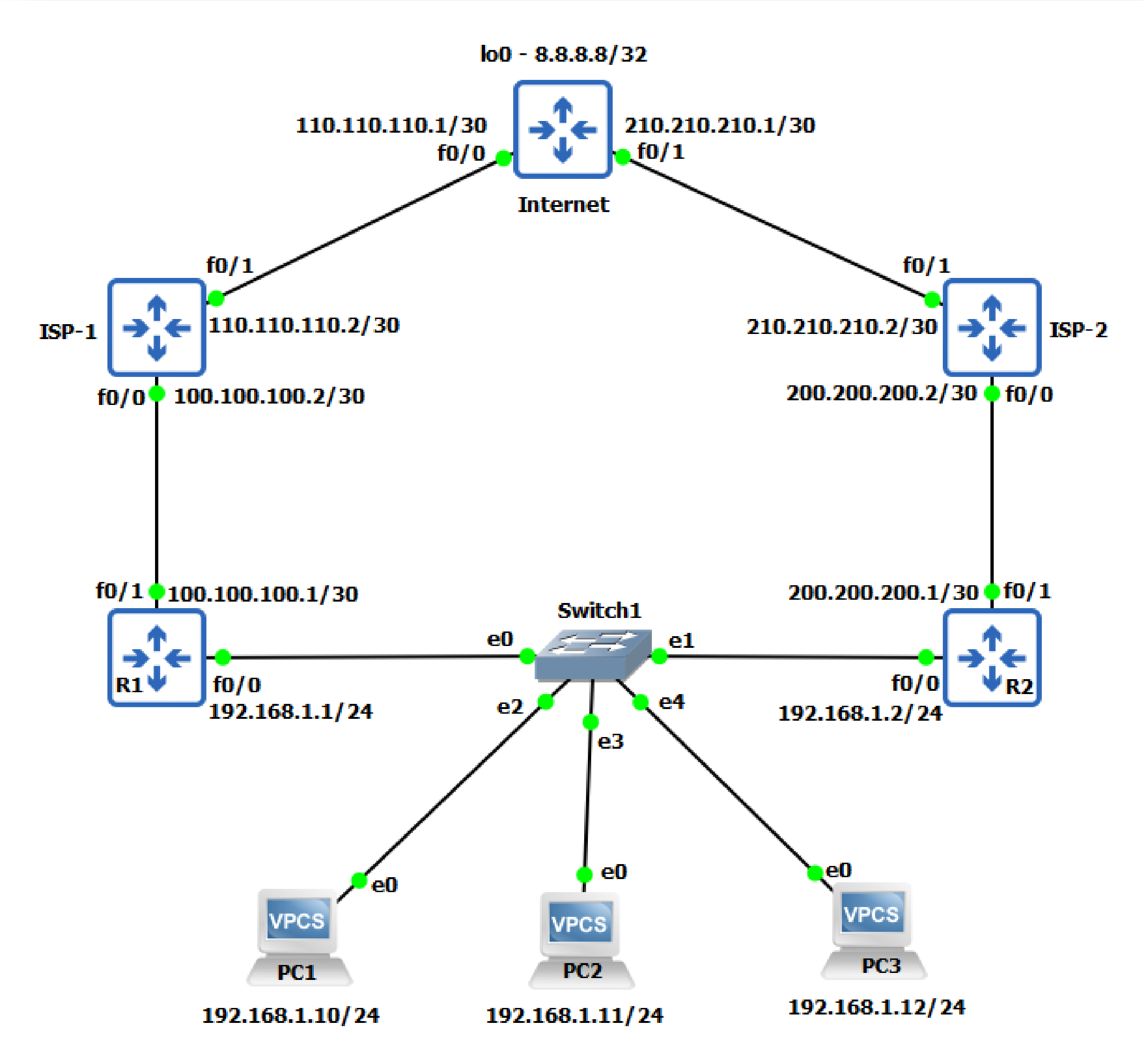
Hot Standby Routing Protocol (HSRP)



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna**.
2. Configure VPCs with IP address and default gateway of 192.168.1.3 (HSRP Virtual IP)
3. Configure default route on R1 & R2 pointing to ISP
4. Configure OSPF on ISP-1, ISP-2 and Internet router.
5. Configure NAT overload (PAT) on R1 & R2
6. Configure HSRP v1 with virtual IP 192.168.1.3 such that R1 will be Active and R2 will be standby.
7. Configure HSRP v2 with virtual IP 192.168.1.3 such that R1 will be Active and R2 will be standby.
8. Configure IP SLA to detect indirect ISP link failure on R1.

info@rsatechforum.in

Task-1: Router basic configuration.

R1 Configuration:

```
R1#config t
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#int fa0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#description Link to LAN
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ip address 100.100.100.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#description Link to ISP-1
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#int fa0/0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#description Link to LAN
R2(config-if)#exit
R2(config)#int fa0/1
R2(config-if)#ip address 200.200.200.1 255.255.255.252
R2(config-if)#no shut
R2(config-if)#description Link to ISP-2
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 1 0
R2(config-line)#exit
R2(config)#exit
R2#
```

ISP-1 Router Configuration:

```
ISP-1#config t
ISP-1(config)#no ip domain-lookup
ISP-1(config)#enable password ccna
ISP-1(config)#int fa0/0
ISP-1(config-if)#ip address 100.100.100.2 255.255.255.252
ISP-1(config-if)#no shut
ISP-1(config-if)#description Link to R1 on fa0/1
ISP-1(config-if)#exit
ISP-1(config)#int fa0/1
ISP-1(config-if)#ip address 110.110.110.2 255.255.255.252
ISP-1(config-if)#no shut
ISP-1(config-if)#description Link to Internet
ISP-1(config-if)#exit
ISP-1(config)#line vty 0 15
ISP-1(config-line)#password cisco
ISP-1(config-line)#login
ISP-1(config-line)#logging synchronous
ISP-1(config-line)#exec-timeout 1 0
ISP-1(config-line)#exit
ISP-1(config)#exit
ISP-1#
```

ISP-2 Router Configuration:

```
ISP-2#config t
ISP-2(config)#no ip domain-lookup
ISP-2(config)#enable password ccna
ISP-2(config)#int fa0/0
ISP-2(config-if)#ip address 200.200.200.2 255.255.255.252
ISP-2(config-if)#no shut
ISP-2(config-if)#description Link to R2 on fa0/1
ISP-2(config-if)#exit
ISP-2(config)#int fa0/1
ISP-2(config-if)#ip address 210.210.210.2 255.255.255.252
ISP-2(config-if)#no shut
ISP-2(config-if)#description Link to Internet
ISP-2(config-if)#exit
ISP-2(config)#line vty 0 15
ISP-2(config-line)#password cisco
ISP-2(config-line)#login
ISP-2(config-line)#logging synchronous
ISP-2(config-line)#exec-timeout 1 0
ISP-2(config-line)#exit
ISP-2(config)#exit
ISP-2#
```

Internet Router Configuration:

Internet#config t
Internet(config)#no ip domain-lookup
Internet(config)#enable password ccna
Internet(config)#int fa0/0
Internet(config-if)#ip address 110.110.110.1 255.255.255.252
Internet(config-if)#no shut
Internet(config-if)#description Link to ISP-1 on fa0/1
Internet(config-if)#exit
Internet(config)#int fa0/1
Internet(config-if)#ip address 210.210.210.1 255.255.255.252
Internet(config-if)#no shut
Internet(config-if)#description Link to ISP-2 on fa0/1
Internet(config-if)#exit
Internet(config)#int lo0
Internet(config-if)#ip address 8.8.8.8 255.255.255.255
Internet(config-if)#exit
Internet(config)#line vty 0 15
Internet(config-line)#password cisco
Internet(config-line)#login
Internet(config-line)#logging synchronous
Internet(config-line)#exec-timeout 1 0
Internet(config-line)#exit
Internet(config)#exit
Internet#

Task-2: Configure VPCs.

PC1 Configuration:

PC1> ip 192.168.1.10/24 192.168.1.3
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.3

PC2 Configuration:

PC2> ip 192.168.1.11/24 192.168.1.3
Checking for duplicate address...
PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.3

PC3 Configuration:

PC3> ip 192.168.1.12/24 192.168.1.3
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.3

Task-3: Configure default route on R1 & R2.

R1 Configuration:

```
R1#config t
R1(config)#ip route 0.0.0.0 0.0.0.0 100.100.100.2
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#ip route 0.0.0.0 0.0.0.0 200.200.200.2
R2(config)#exit
R2#
```

Task-4: Configure OSPF on ISP and Internet routers.

```
ISP-1#config t
ISP-1(config)#router ospf 1
ISP-1(config-router)#network 100.100.100.0 0.0.0.3 area 0
ISP-1(config-router)#network 110.110.110.0 0.0.0.3 area 0
ISP-1(config-router)#exit
ISP-1(config)#exit
ISP-1#
```

```
ISP-2#config t
ISP-2(config)#router ospf 1
ISP-2(config-router)#network 200.200.200.0 0.0.0.3 area 0
ISP-2(config-router)#network 210.210.210.0 0.0.0.3 area 0
ISP-2(config-router)#exit
ISP-2(config)#exit
ISP-2#
```

```
Internet#config t
Internet(config)#router ospf 1
Internet(config-router)#network 110.110.110.0 0.0.0.3 area 0
Internet(config-router)#network 210.210.210.0 0.0.0.3 area 0
Internet(config-router)#network 8.8.8.8 0.0.0.0 area 0
Internet(config-router)#exit
Internet(config)#exit
Internet#
```

✓ Verification & Testing:

R1#sh ip int brief | exclude unassign

```
Interface            IP-Address      OK? Method Status        Protocol
FastEthernet0/0      192.168.1.1     YES manual  up              up
FastEthernet0/1      100.100.100.1   YES manual  up              up
```

R2#sh ip int brief | exclude unassign

```
Interface            IP-Address      OK? Method Status        Protocol
FastEthernet0/0      192.168.1.2     YES manual  up              up
FastEthernet0/1      200.200.200.1   YES manual  up              up
```

ISP-1#sh ip int brief | exclude unassign

```
Interface              IP-Address      OK? Method Status         Protocol
FastEthernet0/0        100.100.100.2   YES manual   up              up
FastEthernet0/1        110.110.110.2   YES manual   up              up
```

ISP-2#sh ip int brief | exclude unassign

```
Interface              IP-Address      OK? Method Status         Protocol
FastEthernet0/0        200.200.200.2   YES manual   up              up
FastEthernet0/1        210.210.210.2   YES manual   up              up
```

Internet#sh ip int brief | exclude unassign

```
Interface              IP-Address      OK? Method Status         Protocol
FastEthernet0/0        110.110.110.1   YES manual   up              up
FastEthernet0/1        210.210.210.1   YES manual   up              up
Loopback0              8.8.8.8         YES manual   up              up
```

R1#sh ip route static

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 100.100.100.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 100.100.100.2
R1#
```

R2#sh ip route static

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 200.200.200.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 200.200.200.2
R2#
```

ISP-1#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      + - replicated route, % - next hop override

Gateway of last resort is not set

     8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/2] via 110.110.110.1, 00:01:00, FastEthernet0/1
     200.200.200.0/30 is subnetted, 1 subnets
O       200.200.200.0 [110/3] via 110.110.110.1, 00:00:50, FastEthernet0/1
     210.210.210.0/30 is subnetted, 1 subnets
O       210.210.210.0 [110/2] via 110.110.110.1, 00:00:50, FastEthernet0/1
ISP-1#
```

ISP-2#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      + - replicated route, % - next hop override

Gateway of last resort is not set

     8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/2] via 210.210.210.1, 00:01:11, FastEthernet0/1
     100.0.0.0/30 is subnetted, 1 subnets
O       100.100.100.0 [110/3] via 210.210.210.1, 00:01:11, FastEthernet0/1
     110.0.0.0/30 is subnetted, 1 subnets
O       110.110.110.0 [110/2] via 210.210.210.1, 00:01:11, FastEthernet0/1
ISP-2#
```

Internet#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      + - replicated route, % - next hop override

Gateway of last resort is not set

     100.0.0.0/30 is subnetted, 1 subnets
O       100.100.100.0 [110/2] via 110.110.110.2, 00:01:16, FastEthernet0/0
     200.200.200.0/30 is subnetted, 1 subnets
O       200.200.200.0 [110/2] via 210.210.210.2, 00:01:06, FastEthernet0/1
```


info@rsatechforum.in

Internet#

Task-5: Configure NAT overload (PAT) on R1 & R2

R1 Configuration:

R1#config t
R1(config)#access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface fa0/1 overload
R1(config)#int fa0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#

R2 Configuration:

R2#config t
R2(config)#access-list 10 permit 192.168.1.0 0.0.0.255
R2(config)#ip nat inside source list 10 interface fa0/1 overload
R2(config)#int fa0/0
R2(config-if)#ip nat inside
R2(config-if)#exit
R2(config)#int fa0/1
R2(config-if)#ip nat outside
R2(config-if)#exit
R2(config)#exit
R2#

Task-6: Configure HSRP v1 with virtual IP 192.168.1.3 such that R1 will be Active and R2 will be standby

R1 Configuration:

R1#config t
R1(config)#int fa0/0
R1(config-if)#standby 1 ip 192.168.1.3
R1(config-if)#standby 1 name HSRPv1
R1(config-if)#standby 1 preempt
R1(config-if)#standby 1 priority 105
R1(config-if)#exit
R1(config)#exit
R1#c

R2 Configuration:

R2#config t
R2(config)#int fa0/0

```
R2(config-if)#standby 1 ip 192.168.1.3
R2(config-if)#standby 1 name HSRPv1
R2(config-if)#exit
R2(config)#exit
R2#
```

- ✓ **Verification & Testing:**

R1#sh standby

```
FastEthernet0/0 - Group 1
  State is Active
    2 state changes, last state change 00:00:56
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.928 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.1.2, priority 100 (expires in 8.688 sec)
  Priority 105 (configured 105)
  Group name is "HSRPv1" (cfgd)
```

R1#sh standby brief

```
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Fa0/0       1    105 P Active  local           192.168.1.2     192.168.1.3
R1#
```

R2#sh standby

```
FastEthernet0/0 - Group 1
  State is Standby
    1 state change, last state change 00:00:35
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.592 secs
  Preemption disabled
  Active router is 192.168.1.1, priority 105 (expires in 10.864 sec)
  Standby router is local
  Priority 100 (default 100)
  Group name is "HSRPv1" (cfgd)
R2#
```

R2#sh standby brief

```
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0       1    100   Standby  192.168.1.1     local           192.168.1.3
```

R2#

PC1> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=255 time=9.324 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=255 time=11.615 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=255 time=5.994 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=255 time=6.519 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=255 time=11.507 ms

PC2> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=255 time=9.682 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=255 time=6.478 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=255 time=4.274 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=255 time=3.403 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=255 time=10.435 ms

PC3> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=255 time=9.244 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=255 time=9.336 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=255 time=5.562 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=255 time=12.519 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=255 time=11.434 ms

PC1> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.327 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=53.359 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=49.293 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=44.325 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=51.393 ms

PC2> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.392 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=54.279 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=52.346 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=44.337 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=49.133 ms

PC3> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.142 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=55.309 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=55.278 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=52.222 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=51.272 ms

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.1:20167	192.168.1.10:20167	8.8.8.8:20167	8.8.8.8:20167
icmp	100.100.100.1:20423	192.168.1.10:20423	8.8.8.8:20423	8.8.8.8:20423
icmp	100.100.100.1:20679	192.168.1.10:20679	8.8.8.8:20679	8.8.8.8:20679
icmp	100.100.100.1:20935	192.168.1.10:20935	8.8.8.8:20935	8.8.8.8:20935
icmp	100.100.100.1:21191	192.168.1.10:21191	8.8.8.8:21191	8.8.8.8:21191

```
icmp 100.100.100.1:21703 192.168.1.11:21703 8.8.8.8:21703   8.8.8.8:21703
icmp 100.100.100.1:21959 192.168.1.11:21959 8.8.8.8:21959   8.8.8.8:21959
icmp 100.100.100.1:22215 192.168.1.11:22215 8.8.8.8:22215   8.8.8.8:22215
icmp 100.100.100.1:22471 192.168.1.11:22471 8.8.8.8:22471   8.8.8.8:22471
icmp 100.100.100.1:22727 192.168.1.11:22727 8.8.8.8:22727   8.8.8.8:22727
icmp 100.100.100.1:23239 192.168.1.12:23239 8.8.8.8:23239   8.8.8.8:23239
icmp 100.100.100.1:23751 192.168.1.12:23751 8.8.8.8:23751   8.8.8.8:23751
icmp 100.100.100.1:24007 192.168.1.12:24007 8.8.8.8:24007   8.8.8.8:24007
icmp 100.100.100.1:24263 192.168.1.12:24263 8.8.8.8:24263   8.8.8.8:24263
icmp 100.100.100.1:24519 192.168.1.12:24519 8.8.8.8:24519   8.8.8.8:24519
R1#
```

R2#sh ip nat translations

```
R2#
```

PC1> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   20.362 ms  9.325 ms  9.506 ms
 2   100.100.100.2   30.284 ms  32.290 ms  32.438 ms
 3   *110.110.110.1   52.202 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC2> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   3.276 ms  9.389 ms  9.539 ms
 2   100.100.100.2   32.416 ms  31.285 ms  31.277 ms
 3   *110.110.110.1   53.197 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   4.653 ms  10.539 ms  9.759 ms
 2   100.100.100.2   31.390 ms  31.397 ms  30.435 ms
 3   *110.110.110.1   52.358 ms (ICMP type:3, code:3, Destination port unreachable)
```

R1#sh arp

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  100.100.100.1           -   ca01.25e0.0006  ARPA   FastEthernet0/1
Internet  100.100.100.2          30   ca03.09c8.0008  ARPA   FastEthernet0/1
Internet  192.168.1.1             -   ca01.25e0.0008  ARPA   FastEthernet0/0
Internet  192.168.1.3             -   0000.0c07.ac01  ARPA   FastEthernet0/0
Internet  192.168.1.10            3   0050.7966.6800  ARPA   FastEthernet0/0
Internet  192.168.1.11            3   0050.7966.6801  ARPA   FastEthernet0/0
Internet  192.168.1.12            2   0050.7966.6802  ARPA   FastEthernet0/0
R1#
```

Test – From PC1 start the PING for 8.8.8.8 and simulate link failure by shutting down the R1's LAN interface.

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#shut
R1(config)#exit
```

```
R1#
```

PC1> ping 8.8.8.8 -c 20

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=52.637 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=46.446 ms
8.8.8.8 icmp_seq=3 timeout
8.8.8.8 icmp_seq=4 timeout
8.8.8.8 icmp_seq=5 timeout
8.8.8.8 icmp_seq=6 timeout
84 bytes from 8.8.8.8 icmp_seq=7 ttl=253 time=78.225 ms
84 bytes from 8.8.8.8 icmp_seq=8 ttl=253 time=45.412 ms
84 bytes from 8.8.8.8 icmp_seq=9 ttl=253 time=50.270 ms
84 bytes from 8.8.8.8 icmp_seq=10 ttl=253 time=52.070 ms
84 bytes from 8.8.8.8 icmp_seq=11 ttl=253 time=53.352 ms
84 bytes from 8.8.8.8 icmp_seq=12 ttl=253 time=50.334 ms
84 bytes from 8.8.8.8 icmp_seq=13 ttl=253 time=55.369 ms
84 bytes from 8.8.8.8 icmp_seq=14 ttl=253 time=47.346 ms
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=52.280 ms
84 bytes from 8.8.8.8 icmp_seq=16 ttl=253 time=48.343 ms
84 bytes from 8.8.8.8 icmp_seq=17 ttl=253 time=49.359 ms
84 bytes from 8.8.8.8 icmp_seq=18 ttl=253 time=46.372 ms
84 bytes from 8.8.8.8 icmp_seq=19 ttl=253 time=45.293 ms
84 bytes from 8.8.8.8 icmp_seq=20 ttl=253 time=47.240 ms

R1#
*Nov 22 11:34:42.167: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Active -> Init
R1#

R2#
*Nov 20 18:57:43.435: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Standby -> Active
R2#
```

R1#sh standby

```
FastEthernet0/0 - Group 1
  State is Init (interface down)
    3 state changes, last state change 00:00:40
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is unknown
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Preemption enabled
  Active router is unknown
  Standby router is unknown
  Priority 105 (configured 105)
  Group name is "HSRPv1" (cfgd)
R1#
```

R2#sh standby

```
FastEthernet0/0 - Group 1
  State is Active
    2 state changes, last state change 00:01:35
```

```
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.072 secs
  Preemption disabled
  Active router is local
  Standby router is unknown
  Priority 100 (default 100)
  Group name is "HSRPv1" (cfgd)
R2#
```

PC1> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.2   9.360 ms  10.436 ms  10.376 ms
 2   200.200.200.2   31.312 ms  31.346 ms  31.264 ms
 3   *210.210.210.1   53.224 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC2> ping 8.8.8.8

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=49.562 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=53.218 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=49.276 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=53.257 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=53.219 ms
```

PC2> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.2   7.703 ms  9.603 ms  10.647 ms
 2   200.200.200.2   30.302 ms  30.413 ms  30.420 ms
 3   *210.210.210.1   53.249 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> ping 8.8.8.8

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=52.128 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=55.628 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=45.410 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=48.274 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=55.263 ms
```

PC3> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.2   7.485 ms  9.517 ms  9.435 ms
 2   200.200.200.2   32.348 ms  30.256 ms  31.366 ms
 3   *210.210.210.1   52.228 ms (ICMP type:3, code:3, Destination port unreachable)
```

Test – Simulate link recovery by making R1's LAN interface up.

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exit
```


info@rsatechforum.in

```
R1#
*Nov 20 19:43:07.683: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Listen -> Active
R1#

R2#
*Nov 20 19:43:19.395: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Standby
R2#
```

R1#sh standby

```
FastEthernet0/0 - Group 1
  State is Active
    4 state changes, last state change 00:00:57
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.984 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.1.2, priority 100 (expires in 8.688 sec)
  Priority 105 (configured 105)
  Group name is "HSRPv1" (cfgd)
R1#
```

R2#sh standby

```
FastEthernet0/0 - Group 1
  State is Standby
    4 state changes, last state change 00:00:23
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.464 secs
  Preemption disabled
  Active router is 192.168.1.1, priority 105 (expires in 8.288 sec)
  Standby router is local
  Priority 100 (default 100)
  Group name is "HSRPv1" (cfgd)
R2#
```

To Remove HSRP

On R1 router:

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#no standby 1
R1(config-if)#exit
R1(config)#exit
R1#
*Nov 22 11:40:43.759: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Active -> Disabled
```

On R2 router:

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#no standby 1
R2(config-if)#exit
R2(config)#exit
R2#
*Nov 22 11:40:50.351: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Active -> Disabled
```

Task-7: Configure HSRP v2 with virtual IP 192.168.1.3 such that R1 will be Active and R2 will be standby.

R1 Configuration:

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#standby version 2
R1(config-if)#standby 10 ip 192.168.1.3
R1(config-if)#standby 10 name HSRPv2
R1(config-if)#standby 10 preempt
R1(config-if)#standby 10 priority 105
R1(config-if)#exit
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#standby version 2
R2(config-if)#standby 10 ip 192.168.1.3
R2(config-if)#standby 10 name HSRPv2
R2(config-if)#standby 10 preempt
R2(config-if)#exit
R2(config)#exit
R2#
```

Task-8: Configure IP SLA to detect indirect ISP link failure on R1.

R1 Configuration:

```
R1#config t
R1(config)#ip sla 1
R1(config-ip-sla)#icmp-echo 8.8.8.8
R1(config-ip-sla-echo)#exit
R1(config)#ip sla schedule 1 start-time now life forever
R1(config)#track 1 ip sla 1 reachability
R1(config-track)#exit
R1(config)#int fa0/0
R1(config-if)#standby 10 track 1 decrement 10
```

```
R1(config-if)#exit
R1(config)#exit
R1#
```

✓ **Verification & Testing:**

R1#sh standby

```
FastEthernet0/0 - Group 10 (version 2)
  State is Active
    2 state changes, last state change 00:02:29
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c9f.f00a
    Local virtual MAC address is 0000.0c9f.f00a (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.760 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.1.2, priority 100 (expires in 10.880 sec)
  Priority 105 (configured 105)
  Group name is "HSRPv2" (cfgd)
R1#
```

R2#sh standby

```
FastEthernet0/0 - Group 10 (version 2)
  State is Standby
    1 state change, last state change 00:18:48
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c9f.f00a
    Local virtual MAC address is 0000.0c9f.f00a (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.192 secs
  Preemption enabled
  Active router is 192.168.1.1, priority 105 (expires in 8.064 sec)
    MAC address is ca01.25e0.0008
  Standby router is local
  Priority 100 (default 100)
  Group name is "HSRPv2" (cfgd)
R2#
```

R1#sh standby brief

```
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Fa0/0       10   105 P Active  local           192.168.1.2     192.168.1.3
R1#
```

R2#sh standby brief

```
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Fa0/0       10   100 P Standby 192.168.1.1     local           192.168.1.3
R2#
```

R1#sh ip sla summary
IPSLAs LaHSRPv1 Operation Summary
Codes: * active, ^ inactive, ~ pending

```
ID      Type        Destination     Stats     Return      Last
                                    (ms)      Code        Run
-----------------------------------------------------------------------
*1      icmp-echo   8.8.8.8         RTT=64    OK          33 seconds ago
```

R1#sh track

```
Track 1
  IP SLA 1 reachability
  Reachability is Up
    2 changes, last change 00:02:33
  LaHSRPv1 operation return code: OK
  LaHSRPv1 RTT (millisecs) 64
  Tracked by:
    HSRP FastEthernet0/0 10
R1#
```

PC1> ping 8.8.8.8

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=78.064 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=78.170 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=52.201 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=49.133 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=47.276 ms
```

PC2> ping 8.8.8.8

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=72.272 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=47.126 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=52.052 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=47.013 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=53.242 ms
```

PC3> ping 8.8.8.8

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=64.368 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=55.144 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=47.282 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=48.136 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=48.162 ms
```

PC1> trace 8.8.8.8

```
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   10.455 ms  10.348 ms  9.310 ms
 2   100.100.100.2   43.039 ms  30.164 ms  31.074 ms
 3   *110.110.110.1   53.173 ms (ICMP type:3, code:3, Destination port unreachable)
```

R1#sh arp

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  100.100.100.1          -    ca01.25e0.0006  ARPA   FastEthernet0/1
Internet  100.100.100.2         23    ca03.09c8.0008  ARPA   FastEthernet0/1
```


info@rsatechforum.in

```
Internet  192.168.1.1          -    ca01.25e0.0008  ARPA   FastEthernet0/0
Internet  192.168.1.3          -    0000.0c9f.f00a  ARPA   FastEthernet0/0
Internet  192.168.1.10         0    0050.7966.6800  ARPA   FastEthernet0/0
Internet  192.168.1.11         0    0050.7966.6801  ARPA   FastEthernet0/0
Internet  192.168.1.12         0    0050.7966.6802  ARPA   FastEthernet0/0
R1#
```

Test – From PC1 start the PING for 8.8.8.8 and simulate link failure by shutting down the ISP-1 fa0/1 interface (indirect interface).

```
ISP-1#config t
ISP-1(config)#int fa0/1
ISP-1(config-if)#shut
ISP-1(config-if)#exit
ISP-1#
```

PC1> ping 8.8.8.8 -c 30

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=57.210 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=48.325 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=45.408 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=47.314 ms
8.8.8.8 icmp_seq=5 timeout
8.8.8.8 icmp_seq=6 timeout
8.8.8.8 icmp_seq=7 timeout
8.8.8.8 icmp_seq=5 timeout
8.8.8.8 icmp_seq=6 timeout
8.8.8.8 icmp_seq=7 timeout
8.8.8.8 icmp_seq=5 timeout
8.8.8.8 icmp_seq=6 timeout
8.8.8.8 icmp_seq=7 timeout
8.8.8.8 icmp_seq=7 timeout
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=107.015 ms
84 bytes from 8.8.8.8 icmp_seq=16 ttl=253 time=63.207 ms
84 bytes from 8.8.8.8 icmp_seq=17 ttl=253 time=52.338 ms
84 bytes from 8.8.8.8 icmp_seq=18 ttl=253 time=54.312 ms
84 bytes from 8.8.8.8 icmp_seq=19 ttl=253 time=50.383 ms
84 bytes from 8.8.8.8 icmp_seq=20 ttl=253 time=44.426 ms
84 bytes from 8.8.8.8 icmp_seq=21 ttl=253 time=55.334 ms
84 bytes from 8.8.8.8 icmp_seq=22 ttl=253 time=44.242 ms
84 bytes from 8.8.8.8 icmp_seq=23 ttl=253 time=45.351 ms
84 bytes from 8.8.8.8 icmp_seq=24 ttl=253 time=50.275 ms
84 bytes from 8.8.8.8 icmp_seq=25 ttl=253 time=47.410 ms
84 bytes from 8.8.8.8 icmp_seq=26 ttl=253 time=45.320 ms
84 bytes from 8.8.8.8 icmp_seq=27 ttl=253 time=46.268 ms
84 bytes from 8.8.8.8 icmp_seq=28 ttl=253 time=52.395 ms
```

```
R1#
*Nov 21 19:30:59.971: %TRACKING-5-STATE: 1 ip sla 1 reachability Up->Down
*Nov 21 19:31:00.679: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Active -> Speak
*Nov 21 19:31:11.039: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Speak -> Standby
R1#
```

```
R2#
*Nov 21 19:30:50.583: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Standby -> Active
R2#
```

R1#sh standby

```
FastEthernet0/0 - Group 10 (version 2)
 State is Standby
  7 state changes, last state change 00:03:05
 Virtual IP address is 192.168.1.3
 Active virtual MAC address is 0000.0c9f.f00a
  Local virtual MAC address is 0000.0c9f.f00a (v2 default)
 Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.432 secs
 Preemption enabled
 Active router is 192.168.1.2, priority 100 (expires in 8.992 sec)
  MAC address is ca02.02b4.0008
 Standby router is local
 Priority 95 (configured 105)
  Track object 1 state Down decrement 10
 Group name is "HSRPv2" (cfgd)
R1#
```

R1#sh track

```
Track 1
 IP SLA 1 reachability
 Reachability is Down
  9 changes, last change 00:03:41
 LaHSRPv1 operation return code: Timeout
 Tracked by:
  HSRP FastEthernet0/0 10
R1#
```

R1#sh ip sla summary

```
IPSLAs LaHSRPv1 Operation Summary
Codes: * active, ^ inactive, ~ pending

ID      Type       Destination    Stats    Return    Last
                                  (ms)     Code      Run
-----------------------------------------------------------------------
*1      icmp-echo   8.8.8.8       -        Timeout   0 seconds ago
```

R2#sh standby

```
FastEthernet0/0 - Group 10 (version 2)
 State is Active
  5 state changes, last state change 00:05:25
 Virtual IP address is 192.168.1.3
 Active virtual MAC address is 0000.0c9f.f00a
  Local virtual MAC address is 0000.0c9f.f00a (v2 default)
 Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.568 secs
 Preemption enabled
```


info@rsatechforum.in

```
  Active router is local
  Standby router is 192.168.1.1, priority 95 (expires in 10.912 sec)
  Priority 100 (default 100)
  Group name is "HSRPv2" (cfgd)
R2#
```

R2#sh arp

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.2             -   ca02.02b4.0008  ARPA   FastEthernet0/0
Internet  192.168.1.3             -   0000.0c9f.f00a  ARPA   FastEthernet0/0
Internet  192.168.1.10            0   0050.7966.6800  ARPA   FastEthernet0/0
Internet  192.168.1.11            0   0050.7966.6801  ARPA   FastEthernet0/0
Internet  192.168.1.12            0   0050.7966.6802  ARPA   FastEthernet0/0
Internet  200.200.200.1           -   ca02.02b4.0006  ARPA   FastEthernet0/1
Internet  200.200.200.2          32   ca04.1438.0008  ARPA   FastEthernet0/1
R2#
```

Test – Simulate link recovery by making ISP-1 fa0/1 interface up.

```
ISP-1#config t
ISP-1(config)#int fa0/1
ISP-1(config-if)#no shut
ISP-1(config-if)#exit
ISP-1#
```

```
R1#
*Nov 21 19:38:00.011: %TRACKING-5-STATE: 1 ip sla 1 reachability Down->Up
*Nov 21 19:38:01.575: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Standby -> Active
R1#
```

```
R2#
*Nov 21 19:37:51.555: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Active -> Speak
*Nov 21 19:38:02.223: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Speak -> Standby
R2#
```

R1#sh standby

```
FastEthernet0/0 - Group 10 (version 2)
  State is Active
    8 state changes, last state change 00:06:48
  Virtual IP address is 192.168.1.3
  Active virtual MAC address is 0000.0c9f.f00a
    Local virtual MAC address is 0000.0c9f.f00a (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.648 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.1.2, priority 100 (expires in 8.992 sec)
  Priority 105 (configured 105)
    Track object 1 state Up decrement 10
  Group name is "HSRPv2" (cfgd)
R1#
```

R2#sh standby
FastEthernet0/0 - Group 10 (version 2)
State is Standby
7 state changes, last state change 00:06:43
Virtual IP address is 192.168.1.3
Active virtual MAC address is 0000.0c9f.f00a
Local virtual MAC address is 0000.0c9f.f00a (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.016 secs
Preemption enabled
Active router is 192.168.1.1, priority 105 (expires in 9.856 sec)
MAC address is ca01.25e0.0008
Standby router is local
Priority 100 (default 100)
Group name is "HSRPv2" (cfgd)
R2#

To Remove HSRPv2 and IP SLA

On R1 router:

R1#config t
R1(config)#int fa0/0
R1(config-if)#no standby 10
R1(config-if)#exit
R1(config)#no ip sla 1
R1(config)#no track 1
R1(config)#exit
R1#
***Nov 22 12:09:57.103: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Standby -> Disabled**

On R2 router:

R2#config t
R2(config)#int fa0/0
R2(config-if)#no standby 10
R2(config-if)#exit
R2(config)#exit
R2(config)#exit
R2#
***Nov 22 12:10:11.863: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 10 state Active -> Disabled**

info@rsatechforum.in

Important Commands:

sh ip int brief | exclude unassign
sh ip route static
sh ip route ospf
sh standby
sh standby brief
sh ip nat translations
sh arp